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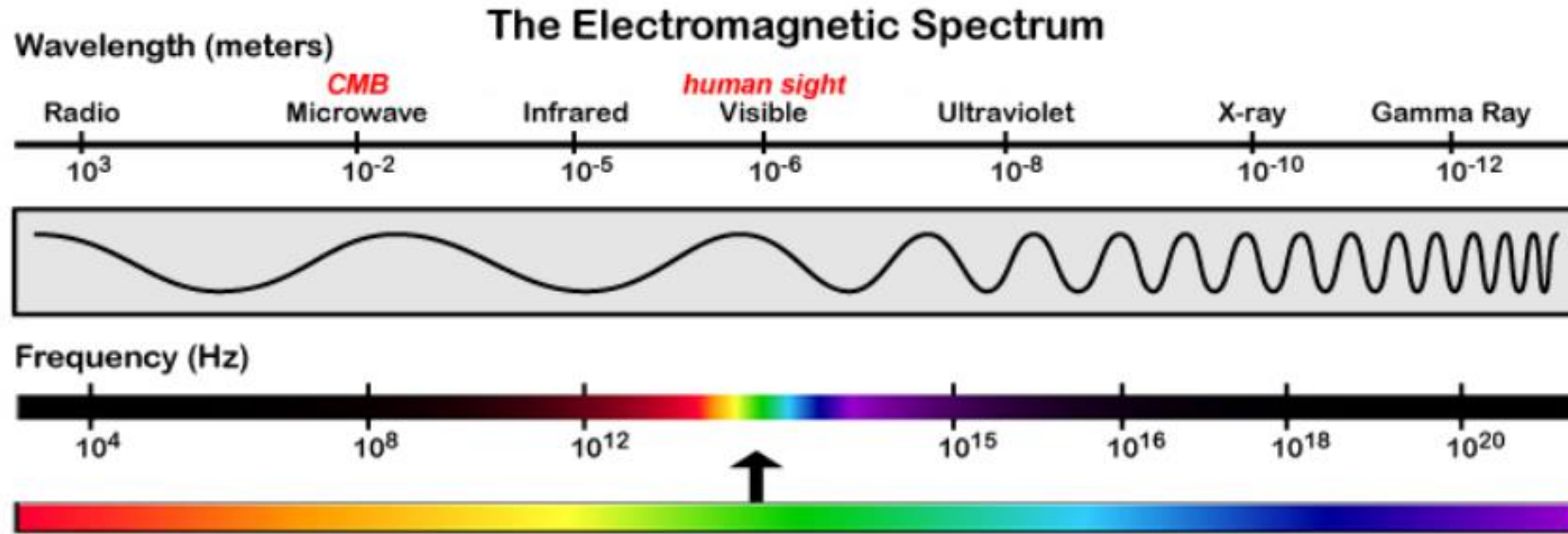
Unit-5

Introduction to Electronic Communication System

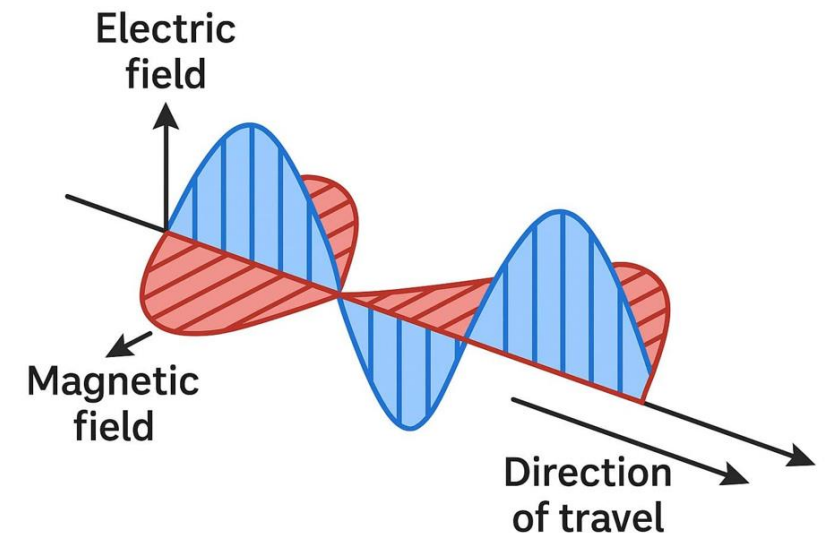


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Electromagnetic Spectrum

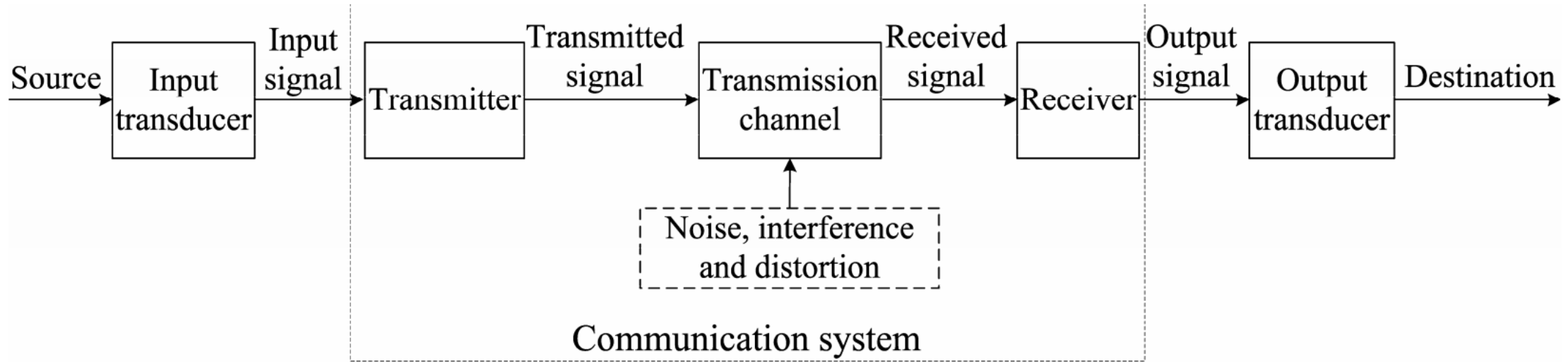


- The Electromagnetic spectrum is the complete range of electromagnetic waves arranged by their frequency and wavelength.
- It includes, in order from longest wavelength to shortest: radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, and gamma rays.
- Each type of wave has unique characteristics and uses.



INTRODUCTION TO COMMUNICATIONS SYSTEMS

Basics of Communication Systems



Input transducer converts the message to an electrical signal.

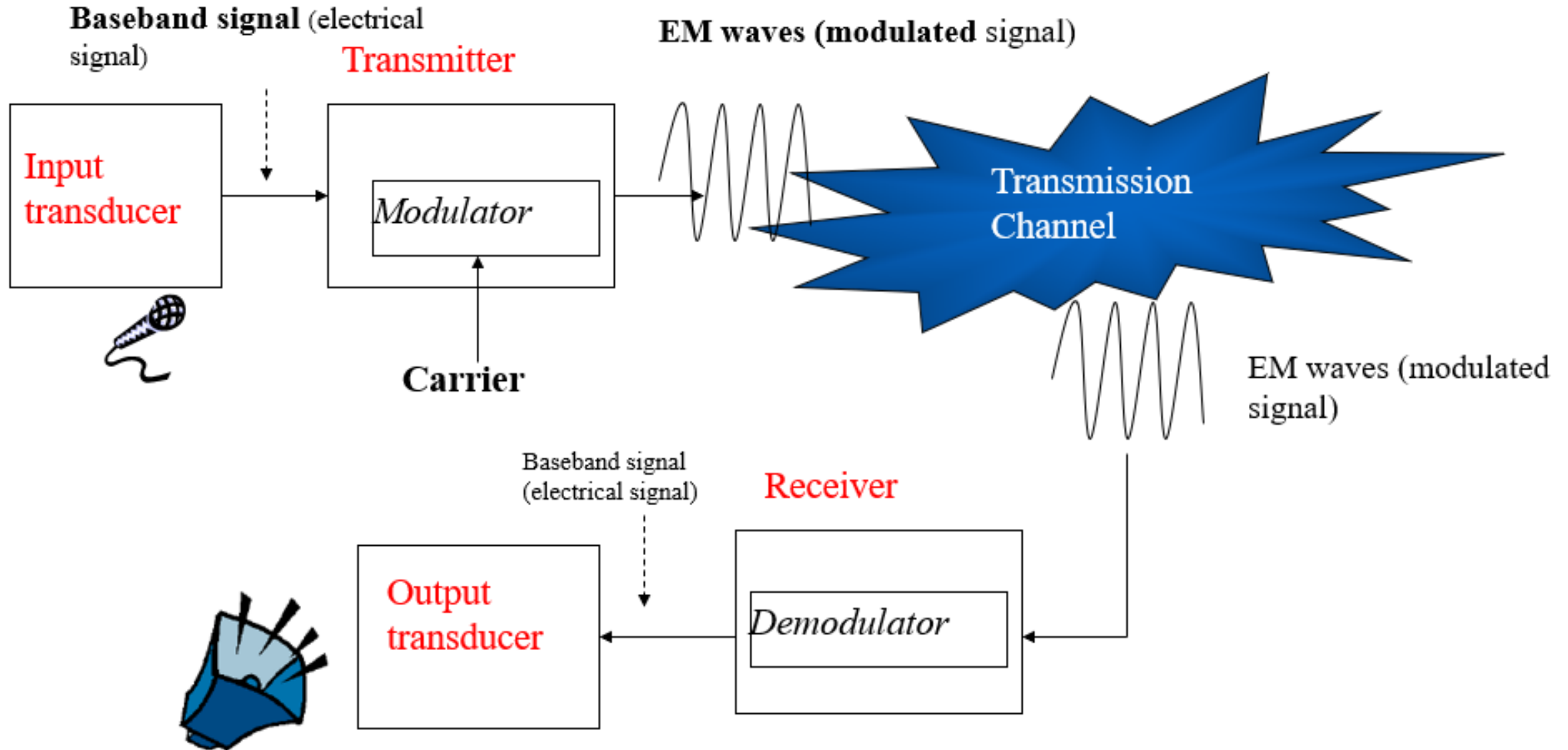
The **transmitter** converts the input signal to transmitted signal suited for the transmission channel.

Transmission channel is the electric medium that bridges the distance from source to destination.

The **receiver** converts the received signal in a form appropriate for the output transducer.

Output transducer converts the output electrical signal to the desired message form.

Basic of communications system



Components of Basic Communication system

▪ **Message:** The communication systems communicates the message. The message comes from the message source

▪ Analog Message: continuous in amplitude and over time

AM, FM for voice sound
Traditional TV for analog video
Record player

▪ Digital message: 0 or 1, or discrete value

Cellular mobile phone
Data on your disk
VCD, DVD

▪ **Transducer:** A **transducer** is a device that converts energy from one form to another, typically transforming physical quantities (like pressure, temperature, or light) into electrical signals (sensors) or vice versa (actuators).

▪ **Types:** Classified as **input transducers** (sensors that convert physical parameters into electrical signals) or **output transducers** (actuators that convert electrical signals into physical action).

▪ **Common Examples:** Microphones (sound to electricity), speakers (electricity to sound), thermometers (temperature to voltage), solar cells (light to electricity), and ultrasonic probes

Transmitter and Receiver

Transmitter: The transmitter is a collection of electronics equipment designed to convert the message signal suitable for transmission over a channel. The typical operations are

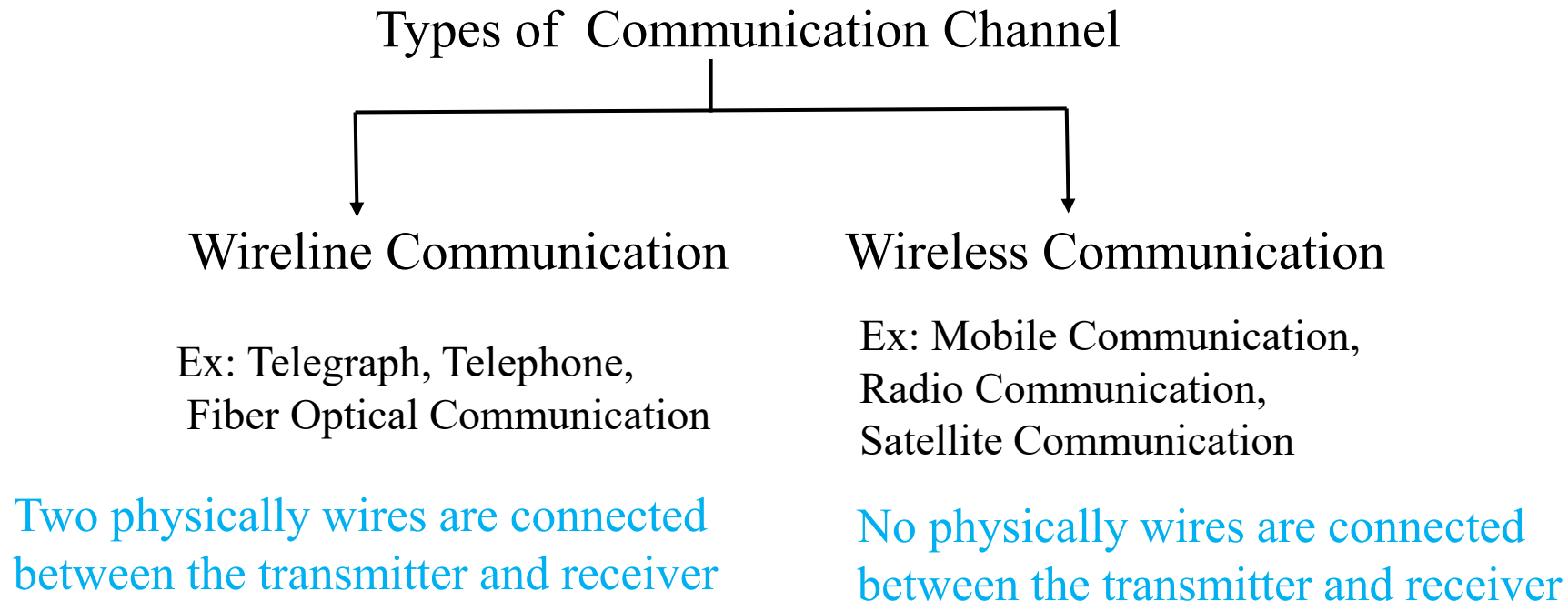
- Encode message data
- Add a carrier signal (modulation)

Receiver : The receiver is a collection of electronics equipment designed to convert the message signal back to its original information. The typical operations are

- Receive signal
- Remove the carrier signal (demodulation)
- Decode the data
- Remove Noise

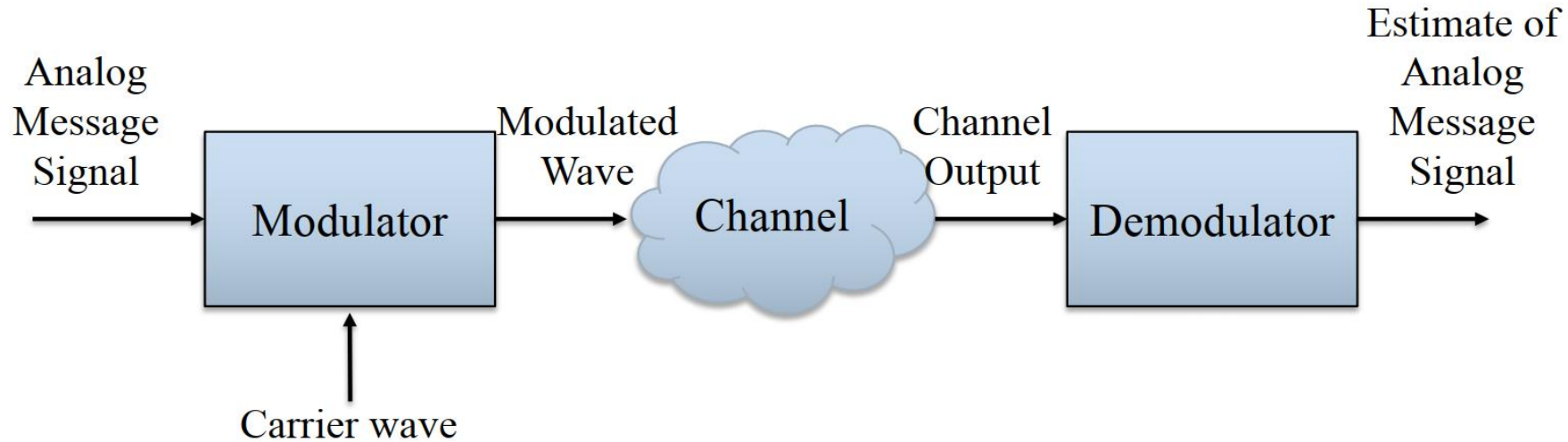
Communication Channel

The communication channel is the medium by which electronics signal is transmitted from one place to another. Ex. Conducting wires, coaxial cable, optical cable, free space



Modulation techniques

Modulation Process



- **Modulation:** Modulation is the process of varying one or more properties of a periodic waveform, called the carrier signal, with a modulating signal that typically contains information to be transmitted.

- **Two signals are used namely**

- $m(t)$ - Message signal/ Modulating signal/ Baseband signal (Low frequency)

- The baseband is used to designate the band of frequencies of the source signal.

- (e.g., audio signal 300Hz-3.4kHz, video 0-5MHz)

- $c(t)$ - Carrier Signal (high frequency)

- Carrier Signal is typically: Sinusoidal or Square Pulse Signal

- Note : After modulation, we get modulated signal $s(t)$

Types of Continuous wave Modulation

❑ Amplitude Modulation (AM)

- ❑ Amplitude modulation is the process of *varying the amplitude of a carrier wave in proportion to the amplitude of a baseband signal*. The frequency of the carrier remains constant

❑ Frequency Modulation (FM)

- ❑ Frequency modulation is the process of *varying the frequency of a carrier wave in proportion to the amplitude of a baseband signal*. The amplitude of the carrier remains constant

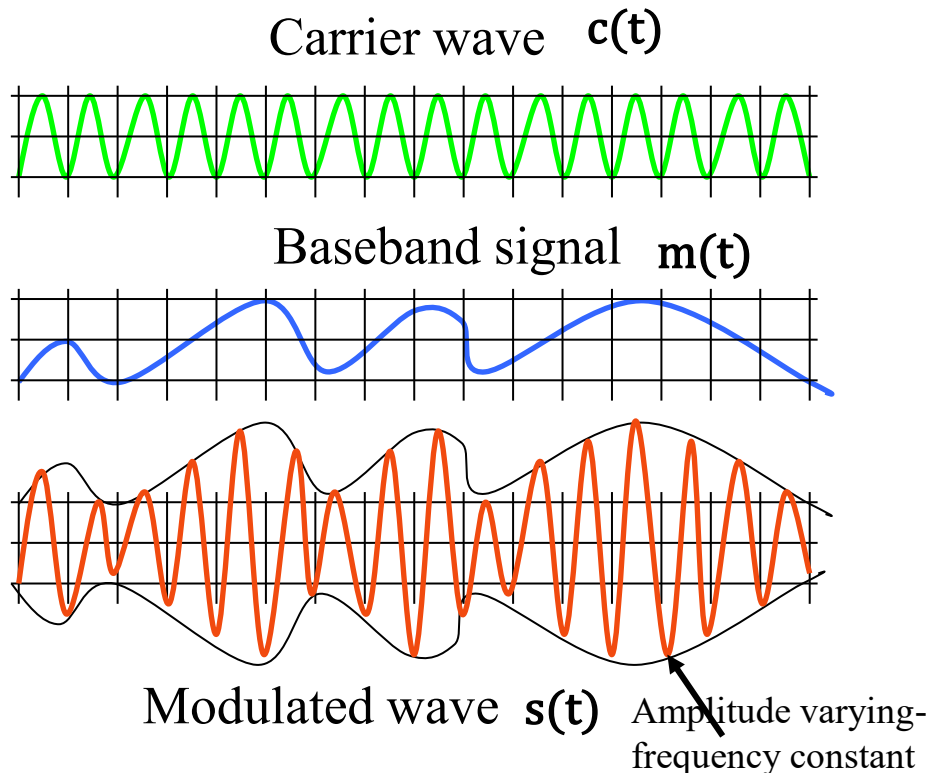
❑ Phase Modulation (PM)

- ❑ Another form of analog modulation technique which we will not discuss

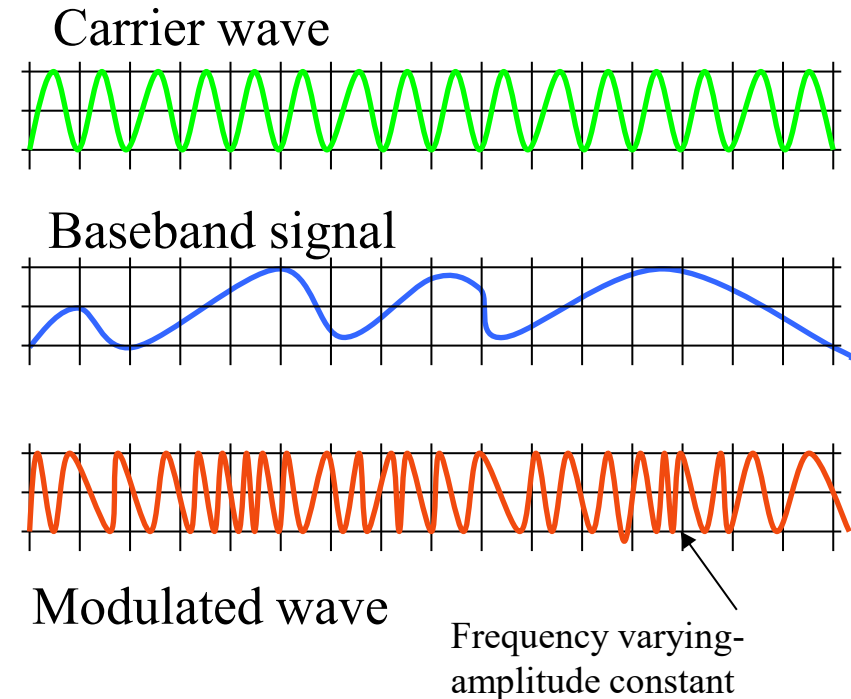
Continuous Wave Modulation

- In analog modulation the basic parameter such as amplitude, frequency or phase of a sinusoidal carrier is varied in proportion to the baseband signal $m(t)$. This results in amplitude modulation (AM) or frequency modulation (FM) or phase modulation (PM).
- The baseband signal $m(t)$ is the modulating signal.
- The sinusoid is the carrier or modulator.

Amplitude Modulation



Frequency Modulation



Needs for modulation

1. Reduction in the height of antenna
2. Increases the range of communication
3. Improves quality of reception
4. Multiplexing is possible

Reduction of antenna height

- For the transmission of radio signals, the antenna height must be of length $\lambda/4$, where λ is the wavelength .

$\lambda = c / f$ where c : is the velocity of light
 f : is the frequency of the signal to be transmitted

- The minimum antenna height required to transmit a baseband signal of $f = 10$ kHz is calculated as follows :

$$\text{Minimum antenna height} = \frac{\lambda}{4} = \frac{c}{4f} = \frac{3 \times 10^8}{4 \times 10 \times 10^3} = 7500 \text{ meters i.e. } 7.5 \text{ km}$$

- The antenna of this height is practically impossible to install .

Long range transmission

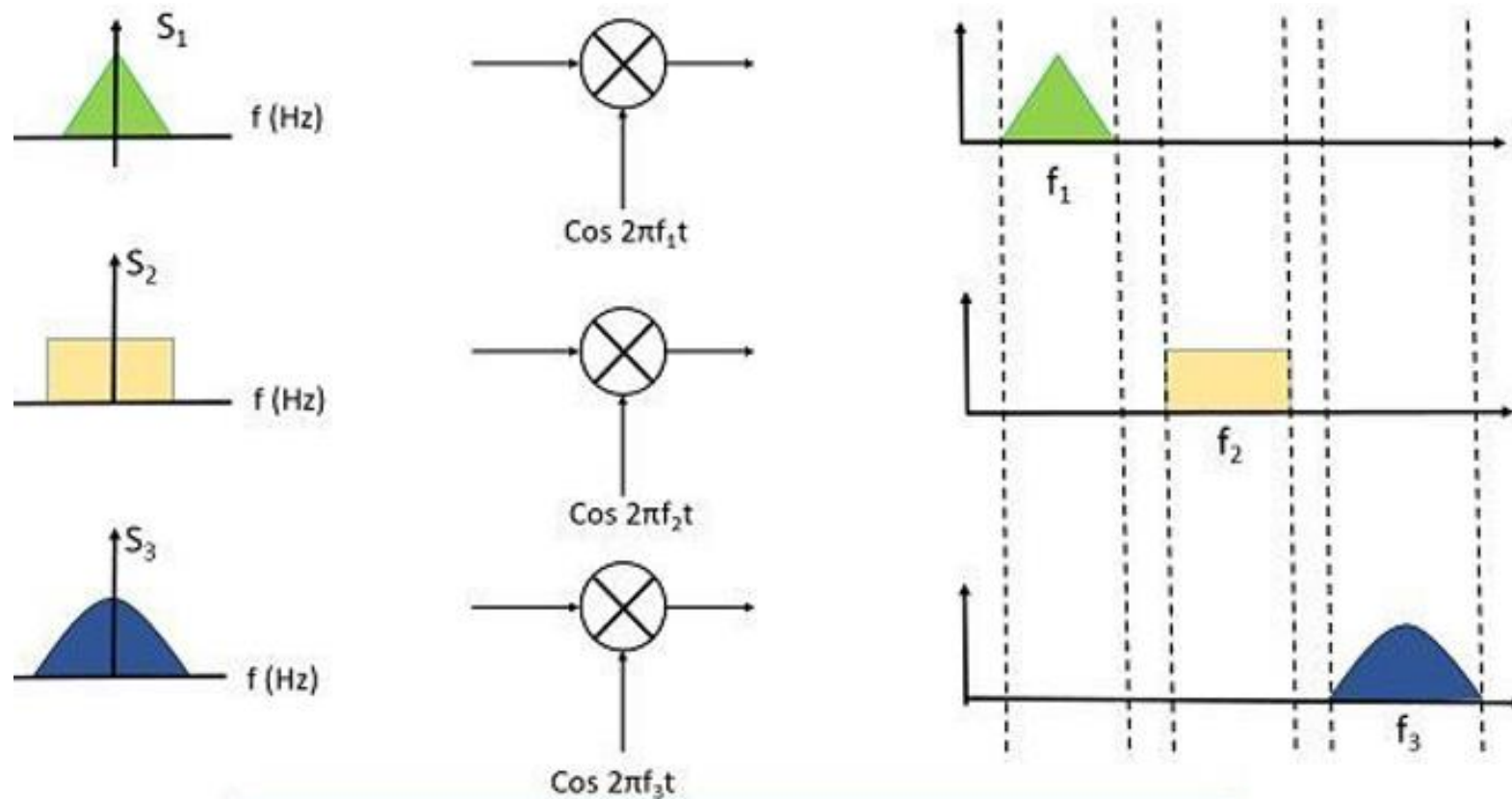
- The frequency of baseband signal is low, and the low frequency signals can not travel long distance when they are transmitted. They get heavily attenuated .
- The modulation process increases the frequency of the signal to be transmitted . Therefore, it increases the range of communication.

Improves Quality of Reception

- With frequency modulation (FM) and the digital communication techniques such as pulse code modulation PCM, the effect of noise is reduced to a great extent .
- This improves quality of reception .

Multiplexing

- Multiplexing is a process in which two or more signals can be transmitted over the same communication channel simultaneously. This is possible only with modulation.



Signals modulated with different Carrier

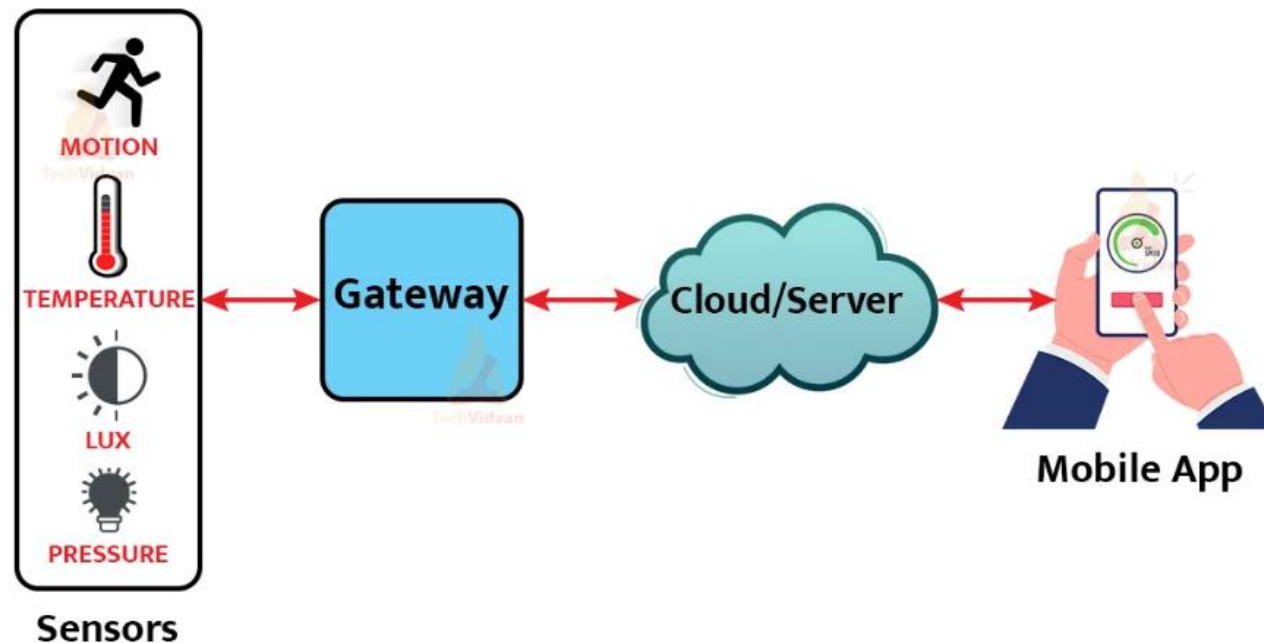
Introduction to Internet of Things (IoT)

What is IoT ?

Internet of Things (IoT) is a system of interrelated, internet-connected objects which are able to collect and transfer data over a wireless network without human intervention.

The Internet of things consists of a **large network of interconnected devices**.

For example, smart fitness bands or watches, driverless cars or drones, smart homes that can be unlocked through smartphones and smart cars, etc.

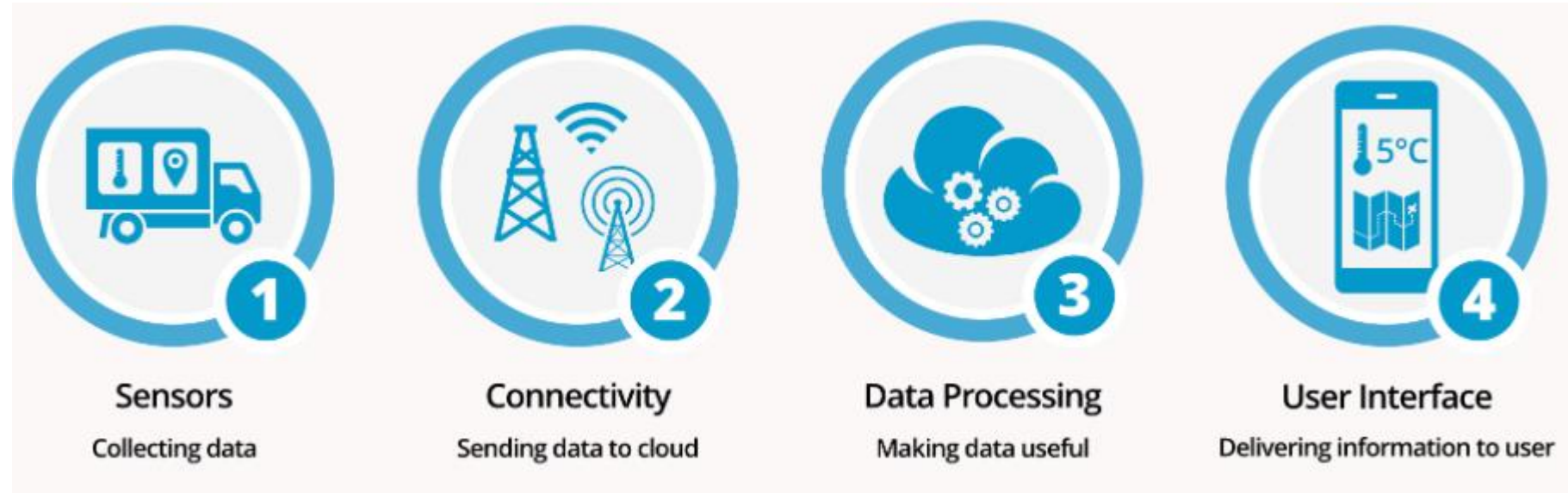


Different Names of IoT

- Internet of Everything
- Smarter Planet
- Machine to Machine (M2M)
- Tsensors (Trillion Sensors)
- The Industrial Internet
- Industry 4.0
- Internet of Things (IoT)

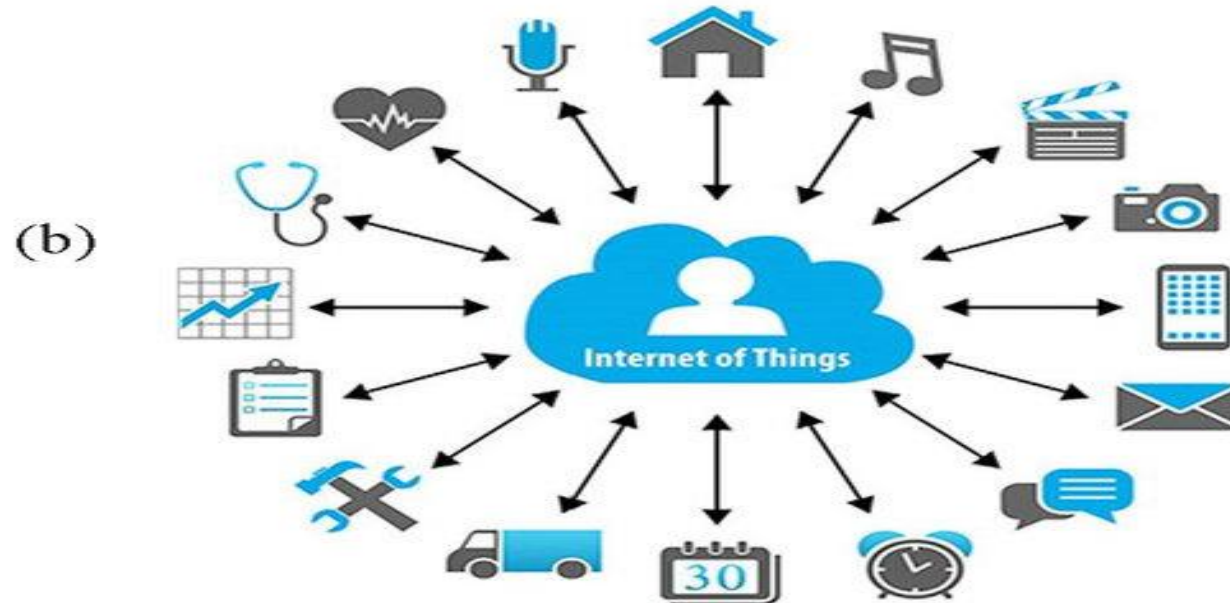
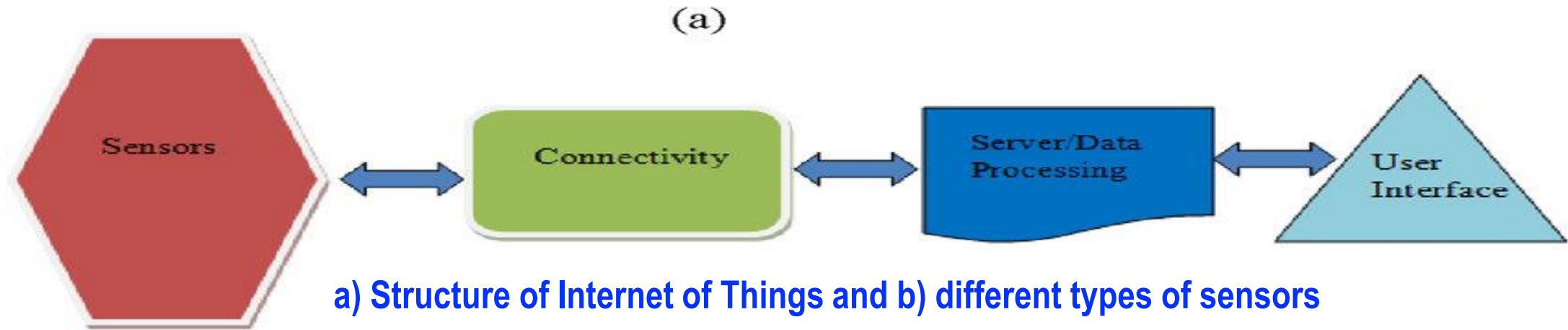
How IoT Works ?

- Sensors/Devices
- Connectivity
- Data Processing
- User Interface



- The sensors are used in the devices and objects and these feed the data to various IoT platforms.
- Further, IoT platforms are used to gather the pinpointed information, detect patterns.
- Thus, with the above process the IoT helps the organizations and institutions in reducing the cost through improved processes efficiency, asset utilization and productivity.

How IoT Works ?



Components of IoT

1. Sensors:

- These devices connect with the external physical environment. They collect the data from the outside changes and store this information.
- A sensor **senses** the changes in the surroundings and notes down these changes. This property makes sensors extremely useful in IoT applications.

For example, there are sensors in your phone such as GPS which track your location and guide you to your destination.

2. Connectivity

- All the collected data is sent to a cloud infrastructure.
- The sensors should be connected to the cloud using various mediums of communications.
- These communication mediums include mobile or satellite networks, Bluetooth, WI-FI, WAN, etc.

Components of IoT

3. Data Processing

- Once that data is collected, and it gets to the cloud, the software performs processing on the gathered data.
- This process can be just checking the temperature, reading on devices like AC or heaters.
- However, it can sometimes also be very complex like identifying objects, using computer vision on video.

4. User Interface

- The information needs to be available to the end-user in some way which can be achieved by triggering alarms on their phones or sending them notification through email or text message.
- The user sometimes might need an interface which actively checks their IoT system.
- For example, the user has a camera installed in his home. He wants to access video recording and all the feeds with the help of a web server.

Advantages of IoT

Advantages of IOT:

As seen from the above example, it is quite obvious IoT if implemented properly can benefit us in this world of globalization as listed below-

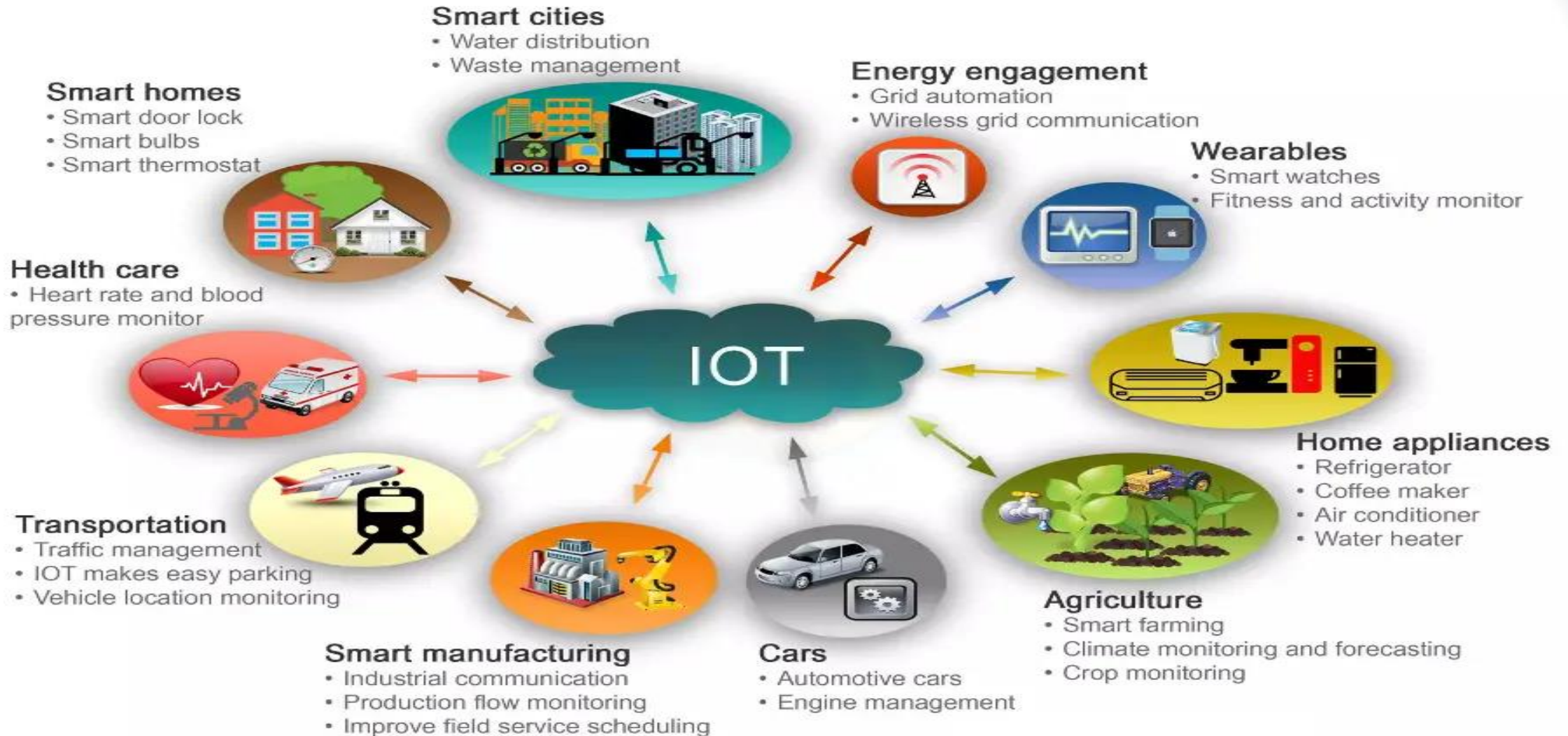
- Eliminates/ reduces the chances of human error
- Decreases wastage of resources- time, material and human efforts
- Boosts confidence & productivity
- Increases security and accuracy
- Enhances the quality

Disadvantages of IoT

Disadvantages of IoT

- If the system is interconnected devices are hacked, entire system security may get jeopardized
- Initial investment on hardware is heavy
- Bringing in IoT will surely absorb many jobs for the work being done manually
- In case of a fault in one device, the entire system may get disrupted
- Too much dependence on IoT, will decrease confidence in human efforts

IoT Applications



1G, 2G, 3G, 4G, 5G

G?

- G → Generation

1G

- Frequency: 150MHz / 900MHz
- Bandwidth: Analog telecommunication (30KHz)
- Characteristic: First wireless communication
- Technology: Analog cellular
- Capacity (data rate): 2kbps
- From 1980 to 1990
- Bad voice quality
- Poor battery, cellphones
- Big cellphones
- Better than nothing, at least its wireless and mobile



2G

- Frequency: 1.8GHz (900MHz), digital telecommunication
- Bandwidth: 900MHz (25MHz)
- Characteristic: Digital
- Technology: Digital cellular, GSM
- Capacity (data rate): 64kbps
- Why better than 1G?

- From 1991 to 2000
- Allows txt msg service
- Signal must be strong or else weak digital signal
- 2.5G
 - 2G cellular technology with GPRS
 - E-Mails
 - Web browsing
 - Camera phones



3G

- Frequency: 1.6 – 2.0 GHz
 - Bandwidth: 100MHz
 - Characteristic: Digital broadband, increased speed
 - Technology: CDMA, UMTS, EDGE
 - Capacity (data rate): 144kbps – 2Mbps
 - Why better than 2G?
- From 2000 to 2010
 - Called smartphones
 - Video calls
 - Fast communication
 - Mobil TV
 - 3G phones rather expensive



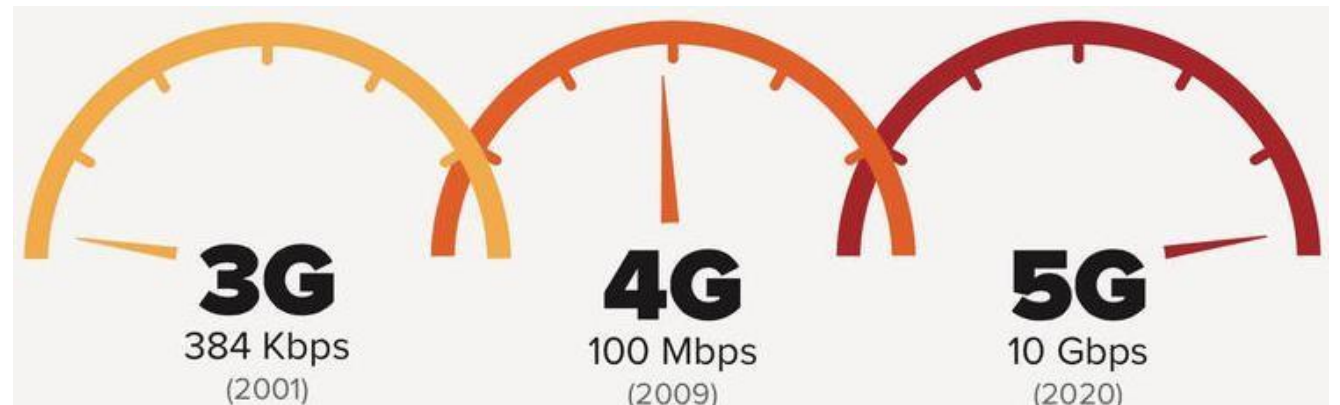
4G

- Frequency: 2 – 8 GHz
 - Bandwidth: 100MHz
 - Characteristic: High speed, all IP
 - Technology: LTE, WiFi
 - Capacity (data rate): 100Mbps – 1Gbps
 - Why better than 3G?
- From 2010 to today (2020?)
 - MAGIC
 - Mobile multimedia
 - Anytime, anywhere
 - Global mobile support
 - Integrated wireless solutions
 - Customized personal service
 - Good QoS + high security
 - Bigger battery usage

5G

- Capacity (data rate): 1Gbps –
ULIMITED?
- Why better than 4G?

- High speed and capacity
- Faster data transmission than 4G
- Supports
 - Interactive multimedia
 - Voice streaming
 - Buckle up.. Internet
- More efficient



	1G	2G	3G	4G	5G
Period	1980 – 1990	1990 – 2000	2000 – 2010	2010 – (2020)	(2020 - 2030)
Frequency Band	850/1900MHz	900/1800/ 1900MHz	900/1800/1900/2100 MHz	1.8- 2.6 GHz	450MHz-6GHz 24.25-52.6GHZ
Bandwidth	Analog signal (30 KHz)	200 KHz (digital)	5 MHz	1.4 - 20 MHz	1000x BW pr unit area
Data rate	2kbps	64kbps	144kbps – 2Mbps	2-100Mbps	More than 1Gbps
Multiplexing	FDMA	TDMA /CDMA	WCDMA	OFDM	Scalable OFDM, NOMA, FBMC
Switching Type	Circuit	Circuit Packet	Circuit Packet	Packet	Packet
Application	Voice	Voice, SMS, Low Data	Voice, Data, Video	High quality Video	IoT based Communication
Technology	Analog cellular	Digital cellular (GSM)	CDMA, UMTS, EDGE	LTE, WiMax, WiFi	Virtual Reality Application

Evolution of mobile phone communications

